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Roll No.

221751

**5th Sem/
Mechanical Engg., Mechanical (Tool & Die)
Subject : CNC Machines & Automation**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 What does PLC stand for?
- a) Process Logic Controller
 - b) Programmable Logic Controller
 - c) Programmed Linear Code
 - d) Power Line Control
- Q.2 Which of the following is an example of a motion system in CNC?
- a) Encoder
 - b) Servo motor
 - c) Transducer
 - d) Trachometer
- Q.3 What is the purpose of a part program in CNC?
- a) To store electrical signals
 - b) To control tool movement
 - c) To lubricate components
 - d) To increase spindle speed

- Q.4 Which type of actuator is commonly used in CNC machines?
- a) Pneumatic b) Hydraulic
 - c) Electric d) All of the above
- Q.5 What is the primary function of a coordinate measuring machine (CMM)?
- a) Measure dimensions
 - b) Cut metal
 - c) Cool the system
 - d) Provide power supply
- Q.6 What is an advantage of Direct Numerical Control (DNC)?
- a) Limited storage
 - b) Centralized Control
 - c) Increased manual effort
 - d) Slower operation

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 Define the term "adaptive control" in CNC machines.
- Q.8 NC machines use binary coding for operation. (True/False)
- Q.9 Name two types of cutting tools used in CNC machining.

- Q.10 ____ is the main advantage of using automation in manufacturing.
- Q.11 What is the role of AGV in flexible manufacturing systems?
- Q.12 Automated inspection improves product quality in CNC manufacturing. (True/False)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Explain the working principle of a servo motor in CNC machines.
- Q.14 What are the advantages of different types of CNC controllers?
- Q.15 Discuss the role of sensors in feedback control of CNC machines.
- Q.16 What are the different types of automation systems used in industries?
- Q.17 Explain the function of tool magazines used in CNC machines.
- Q.18 How does a PLC function in CNC operations?
- Q.19 Describe the use of Robotics in modern CNC manufacturing.
- Q.20 Explain any four G codes.
- Q.21 What are the common faults in CNC machines and their solutions?
- Q.22 Discuss the role of AI in manufacturing automation.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain the classification of NC systems based on motion and feedback systems.

Q.24 Discuss the applications of Flexible Manufacturing Systems (FMS)) in modern industries.

Q.25 Write a finishing cut program of step turning as shown in fig. 1. All dimensions are in mm.

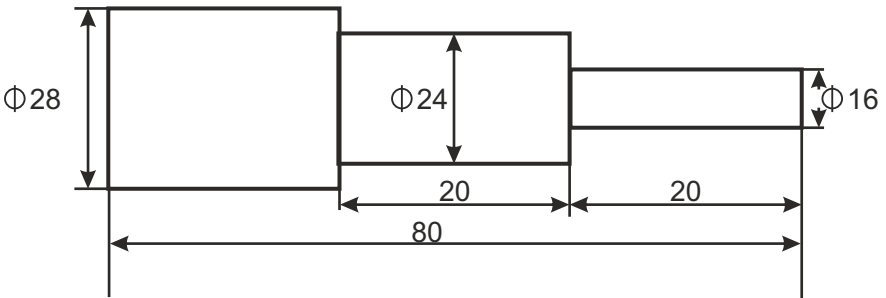


Fig. 1